

Initial Project Planning Template

Date	1 July 2026
Team ID	xxxxxx
Project Name	OptiCrop – Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Product Backlog, Sprint Schedule, and Estimation (4 Marks)

Use the below template to create a product backlog and sprint schedule

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members	Sprint Start Date	Sprint End Date (Planned)
Sprint-1	Smart Crop Recommendation	USN-1	As a farmer, I can enter my soil nutrient values (Nitrogen, Phosphorous, Potassium) and pH so that the system can analyze my land's fertility.	2	High	Bhavani Dunna	28 June 2026	30 June Mar 2026
Sprint-1		USN-2	As a farmer, I can enter local weather metrics (temperature, humidity, rainfall) so the engine can factor in climate conditions for my plot.	1	High	Mondam Sandhya	28 June 2026	30 June 2026
Sprint-1		USN-3	As a farmer, I can submit my soil and weather inputs and receive the top recommended crop for maximum expected yield.	2	High	Mondam Spandhana	28 June 2026	30 June 2026
Sprint-1	Model & Data Pipeline	USN-4	As a system, the cleaned dataset should be used to train and compare multiple ML models, saving the best-performing model as a Joblib file.	3	High	Bhavani Dunna	28 June 2026	30 June 2026
Sprint-2	Crop Suitability & Environmental Assessment	USN-5	As a user, I can select a specific crop (e.g., Coffee) and compare it against my current soil and weather inputs to check compatibility.	2	Medium	Mondam Sandhya	29 June 2026	1 July 2026
Sprint-2		USN-6	As a user, I receive a risk alert when my soil or weather conditions fall outside the safe range required for the selected crop.	1	Medium	Mondam Spandhana	29 June 2026	1 July 2026

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Sprint-2	Agricultural Research & Policy Planning	USN-7	As a researcher, I can adjust regional soil parameters to simulate how crop suitability shifts across an area, supporting food-security planning.	3	Low	Bhavani Dunna	29 June 2026	1 July 2026
Sprint-2	Deployment	USN-8	As a user, I can access the crop recommendation form through a simple HTML/CSS/JS frontend served by the Flask web application.	2	Medium	Mondam Sandhya	29 June 2026	1 July 2026